Experiment 0

Aim : Introduction to Python libraries: NumPy, Pandas, Matplotlib, Seaborn

Data Source : (Student Performance Dataset)

- Type of Data: Structured tabular dataset (CSV format)
- File Format: `.csv` (Comma Separated Values)
- Number of Records (Rows): 20
- Number of Attributes (Columns): 5
- Nature of Dataset: Academic performance dataset (student performance data)
- Data Type: Quantitative (Numerical – Integer values)

Data Description :

The dataset contains information about students' study habits and academic performance.

Column Name	Description	Data Type	Meaning
Hours_Studied	Number of hours studied	Integer	Independent variable
Attendance	Attendance percentage	Integer (%)	Class participation level
Assignment_Score	Marks obtained in assignments	Integer	Internal assessment score
Midterm_Score	Marks obtained in midterm exam	Integer	Mid-semester performance
Final_Score	Marks obtained in final exam	Integer	Final academic outcome

Theory :
1) NumPy (Numerical Python) :

NumPy is the foundational library for numerical computing in Python. It provides a powerful N-dimensional array object (ndarray) and a large collection of mathematical functions to operate on these arrays efficiently.

Why NumPy?

- Faster than Python lists (implemented in C)
- Supports vectorized operations (no explicit loops needed)
- Efficient memory usage
- Basis for libraries like Pandas, SciPy, TensorFlow

Core Concepts

1. **ndarray**
    - Homogeneous (same data type)
    - Supports multi-dimensional data (1D, 2D, 3D, …)
2. **Vectorization**
    - Apply operations to entire arrays at once
    - Example: `a + b` adds two arrays element-wise
3. **Broadcasting**
    - Automatically adjusts array shapes for arithmetic operations
4. **Linear Algebra Support**
    - Matrix multiplication
    - Determinant, inverse, eigenvalues
5. **Random Module**
    - Random sampling, normal distribution, etc.

Applications

- Scientific computing
- Machine Learning
- Signal processing
- Image processing

2) Pandas

Pandas is a powerful library for data manipulation and analysis. It provides flexible data structures to work with structured data.

Why Pandas?

- Easy handling of tabular data (like Excel or SQL tables)
- Built-in data cleaning tools
- Handles missing data efficiently
- Fast and flexible

Core Data Structures

1. **Series**
    - 1D labeled array
    - Like a single column in a table
2. **DataFrame**
    - 2D labeled data structure
    - Rows and columns (like Excel sheet)

Important Features

- Data selection & filtering
- Handling missing values (`NaN`)
- Grouping & aggregation (`groupby`)
- Merging & joining datasets
- Reading/writing data (CSV, Excel, SQL)

Applications

- Data analysis
- Business analytics
- Data preprocessing in ML
- Financial analysis

3) Matplotlib

Matplotlib is a comprehensive library for data visualization in Python. It allows users to create static, animated, and interactive plots.

Why Matplotlib?

- Highly customizable
- Works well with NumPy & Pandas
- Industry-standard visualization library

Basic Plot Types

- Line Plot
- Bar Chart
- Histogram

- Scatter Plot
- Pie Chart

Core Components

1. **Figure** – Entire canvas
2. **Axes** – Plot area
3. **Labels & Legends**
4. **Grid & Styles**

Applications

- Scientific research
- Academic presentations
- Business dashboards
- Data storytelling

4) Seaborn

Seaborn is a statistical data visualization library built on top of Matplotlib. It provides a high-level interface for drawing attractive and informative statistical graphics.

Why Seaborn?

- Beautiful default styles
- Less code than Matplotlib
- Built-in statistical plots
- Works directly with Pandas DataFrames

Popular Plots

- Heatmap
- Boxplot
- Violin plot
- Pairplot
- Regression plot

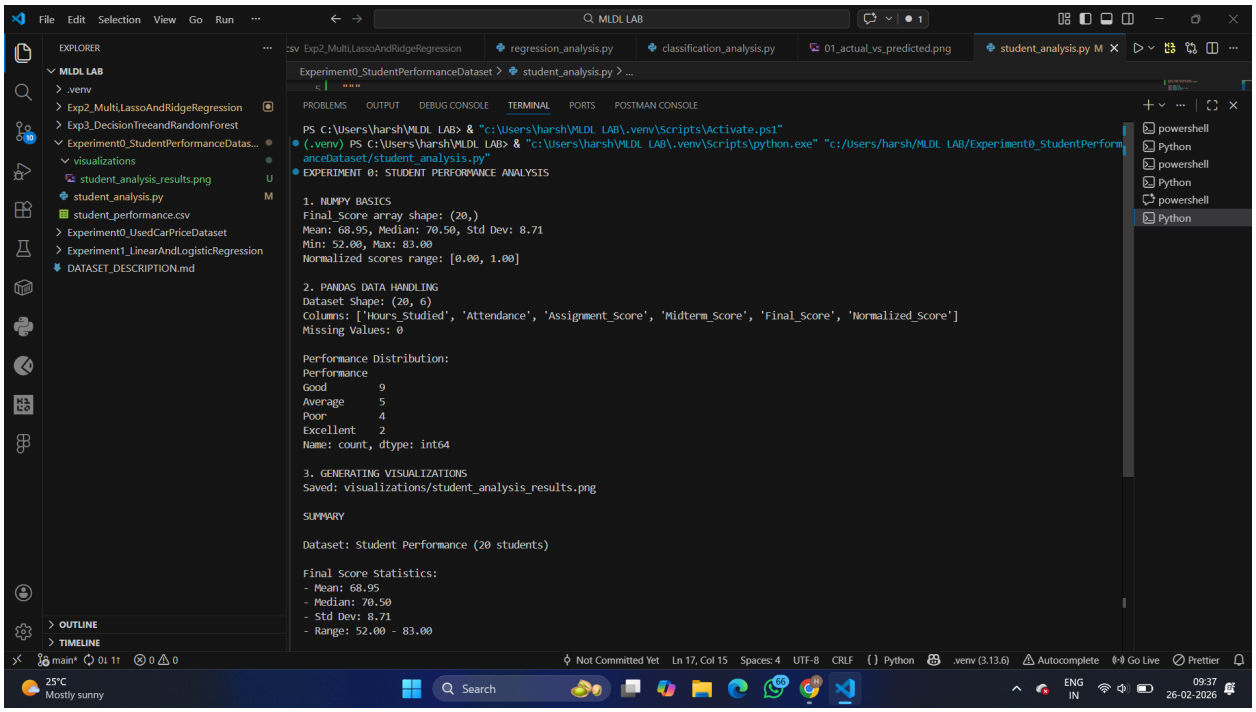
Key Advantages

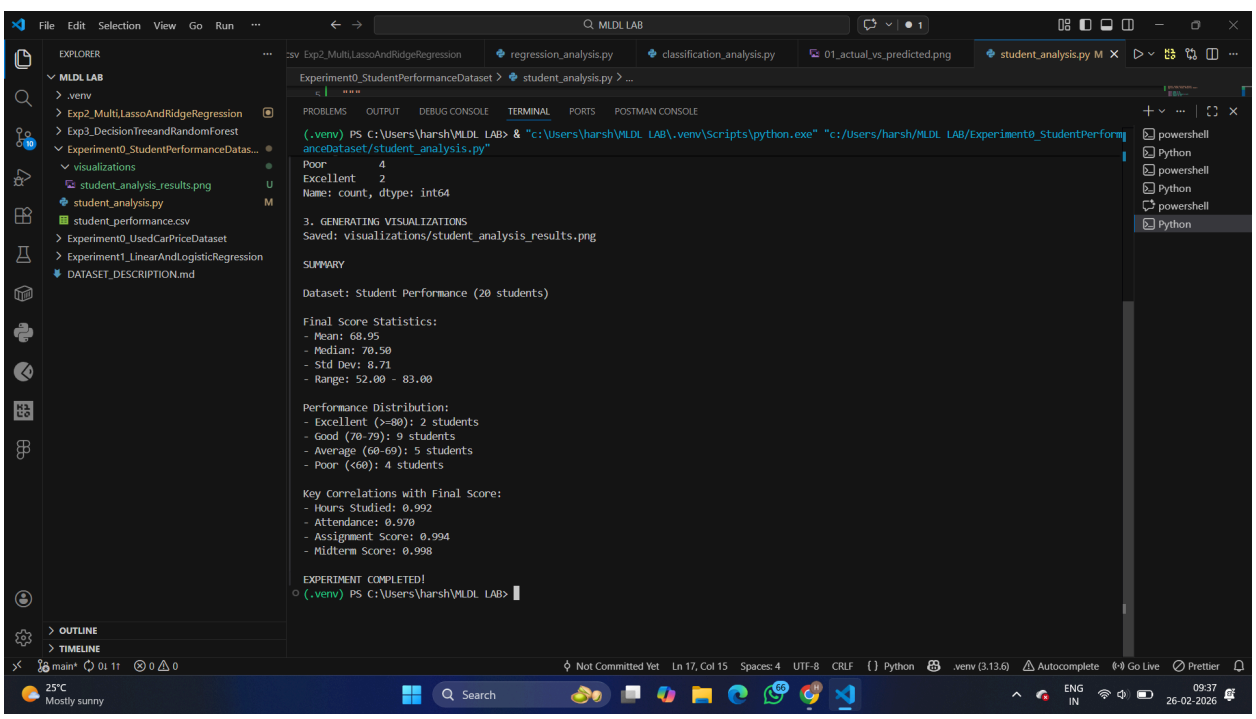
- Automatic aggregation
- Handles categorical data easily
- Built-in themes and color palettes

Applications

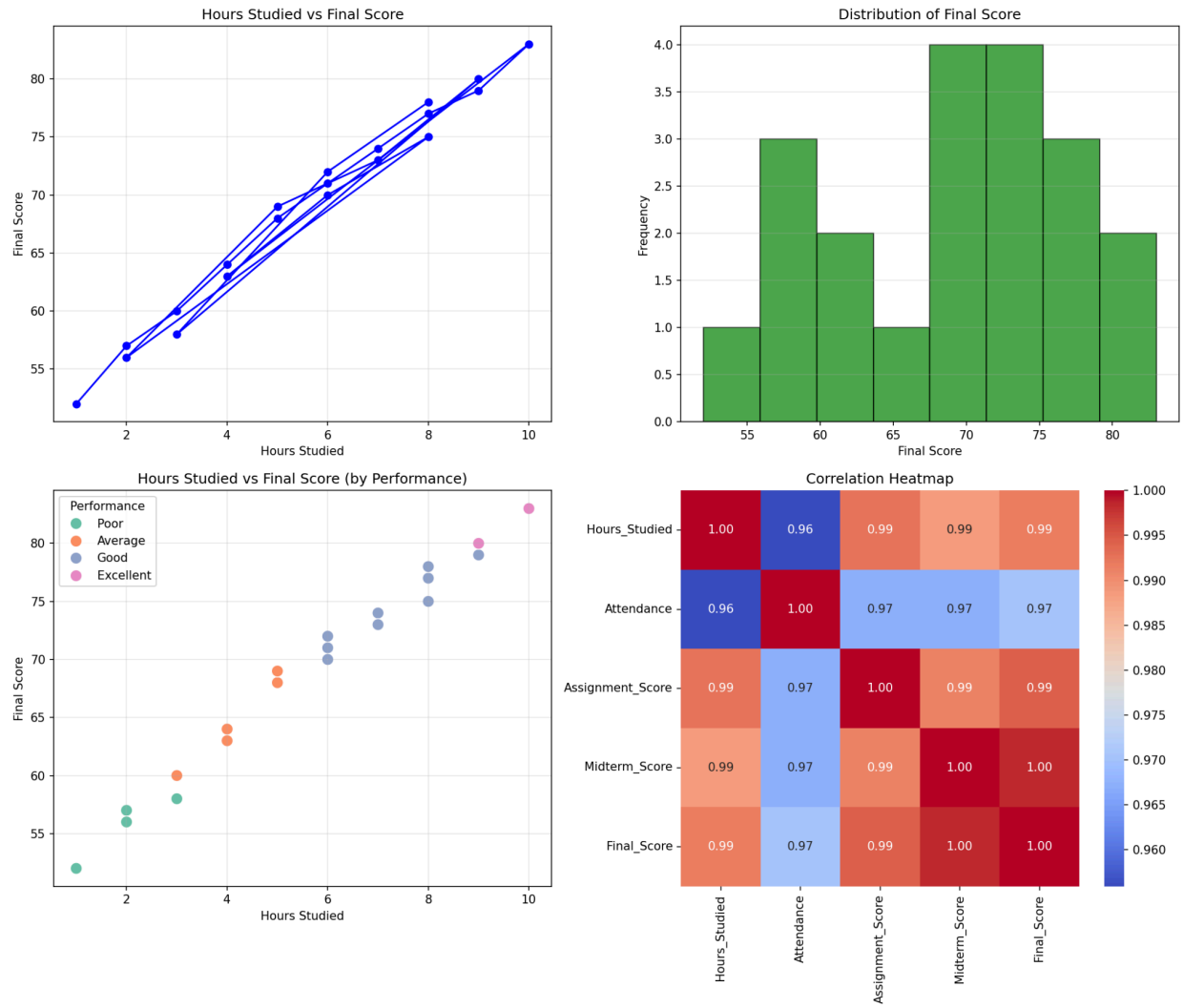
- Exploratory Data Analysis (EDA)
- Statistical analysis
- Machine Learning data visualization

Code :





Output :



Conclusion : In this experiment, we studied the fundamental Python libraries — NumPy, Pandas, Matplotlib, and Seaborn — and applied them to analyze a Student Performance dataset. The dataset was a structured tabular file in CSV format containing 20 records and 5 numerical attributes representing students' academic performance.